

## Choose the Correct Maxwell-Boltzmann Distributions

Subject & topics: Chemistry | Physical | Kinetics Stage & difficulty: A Level P1

Maxwell-Boltzmann distributions show the proportion of molecules with particular speeds. Below are five different Boltzmann diagrams, showing distributions at two temperatures.

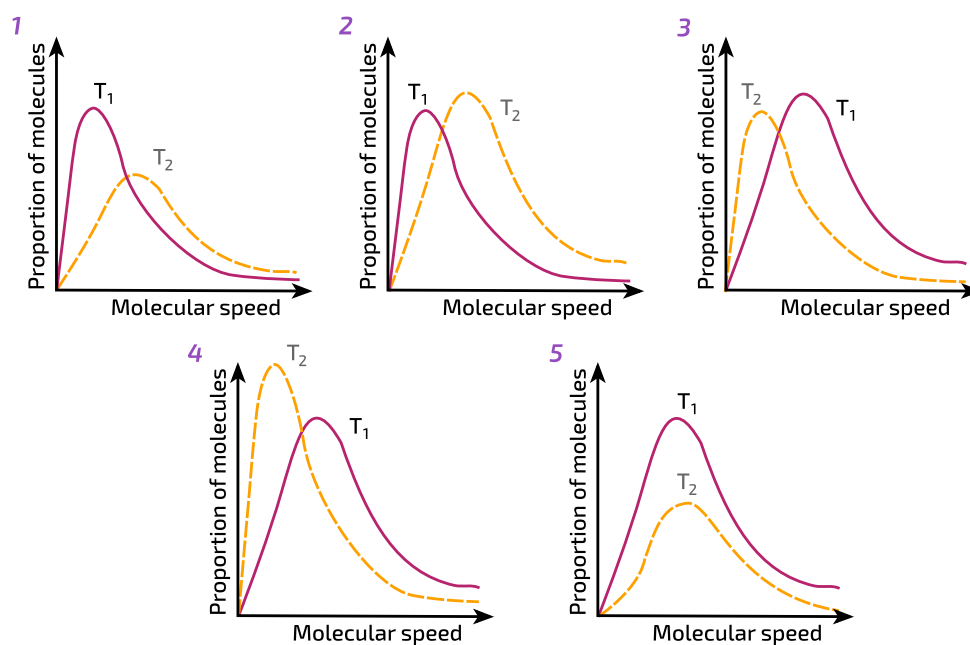


Figure 1: Five Boltzmann distributions.

Which one of the above diagrams correctly represents the Boltzmann distribution of molecular speeds at two temperatures,  $T_1$  and  $T_2$ , where  $T_2 > T_1$ ?

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5

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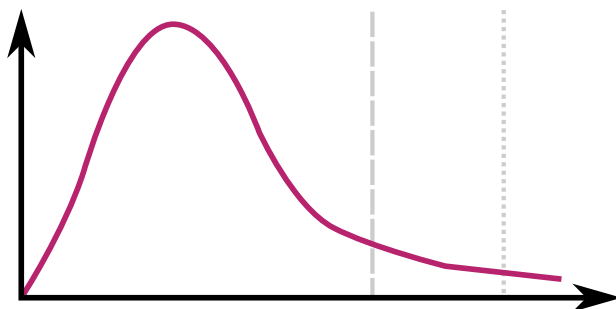
## Examining the Maxwell-Boltzmann Distribution

**Subject & topics:** Chemistry | Physical | Kinetics    **Stage & difficulty:** A Level P1

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This question will examine the features of the Maxwell-Boltzmann distribution, and what happens to it under various conditions.

Below is a diagram of a Maxwell-Boltzmann distribution at an arbitrary temperature (with two vertical lines that should be ignored for now).



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**Figure 1:** Maxwell-Boltzmann Distribution at an arbitrary temperature

## Part A

### Labelling the axes

The axes on the diagram above are not labelled. What would be a suitable label for the y-axis? Choose the most appropriate.

- ☐ Time
- ☐ Kinetic energy
- ☐ Number of particles
- ☐ Cumulative number of particles
- ☐ Molecular speed
- ☐ Molecular velocity

And what would be a suitable label for the x-axis?

- ☐ Time
- ☐ Proportion of particles
- ☐ Number of particles
- ☐ Molecular speed
- ☐ Molecular velocity
- ☐ Distance

### Part B

#### Activation energy

The two vertical lines represent the point at which the particles reach the activation energy; one line for when the reaction was carried out in the presence of a catalyst, and one without catalyst.

Which line represents the activation energy when carried out in the presence of a catalyst?

- ☐ The dashed line
- ☐ The dotted line

How does this show that adding a catalyst increases the rate of reaction?

### Part C

#### Effect of temperature

If the temperature changes, the shape of the plot changes.

If the temperature was **increased**, in which direction would the **peak** of the distribution move?

- ☐ It would not move
- ☐ It depends on the initial temperature
- ☐ It depends on other factors
- ☐ To the right
- ☐ To the left

#### Part D

##### Width of the peak

What is the general change to the width of the peak at a higher temperature?

- ☐ The width would not change
- ☐ It depends on the initial temperature
- ☐ It depends on other factors
- ☐ The peak narrows
- ☐ The peak becomes broader

#### Part E

##### Lowered temperature

If the temperature was **lowered**, what would happen to the height of the peak?

- ☐ It would not change
- ☐ It depends on the initial temperature
- ☐ It depends on other factors
- ☐ It would increase
- ☐ It would decrease

Part F

Constant feature

Which feature of the graph stays constant, regardless of changing temperature?

- ☐ The area under the graph
- ☐ The height of the peak
- ☐ The gradient at the origin
- ☐ The position of the peak
- ☐ The mean molecular speed

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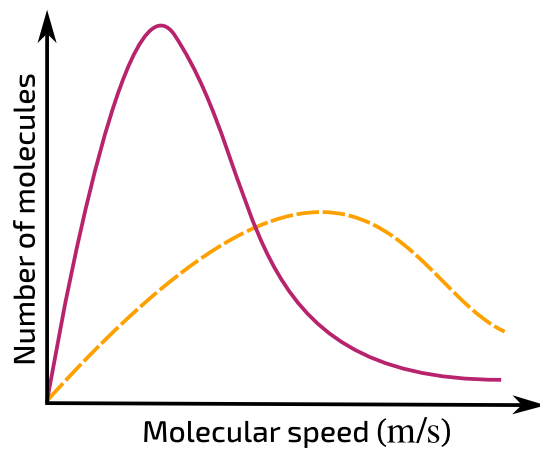
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## Two Boltzmann Distributions

**Subject & topics:** Chemistry | Physical | Kinetics    **Stage & difficulty:** A Level P1

This question investigates the effect of changing temperature on the Maxwell-Boltzmann distribution.



**Figure 1:** Maxwell-Boltzmann graph at two temperatures.

### Part A

#### Temperature

Which line shows the distribution at a higher temperature?

- ☐ The line with the lower peak.
- ☐ The line with the higher peak.



## Part B

### Extending the distribution

What would be seen if the plots were extended to a very large molecular speed?

- ☐ The plots will reach zero, the one at lower temperature first
- ☐ The plots will tend to the same non-zero value
- ☐ The plots will each tend to different non-zero values
- ☐ The plots will tend to zero
- ☐ The plots will reach zero, the one at higher temperature first

## Part C

### Effect on rate

It is suggested that there is an arbitrary molecular speed at which collisions between particles will result in a chemical reaction. Given this, what is the effect of a temperature increase on the rate of reaction?

- ☐ At higher temperature, the particles will collide more often, but with the same energy, so the rate of reaction will increase.
- ☐ At higher temperature, the particles will not react, so the rate will decrease.
- ☐ At a higher temperature, more particles will have sufficient speed and hence energy to react, so the rate of reaction will increase.
- ☐ At higher temperature, the particles will collide with too much energy, so the rate of reaction will decrease.

**Part D**

**Naming**

For the particle with the energy described above, what is this energy commonly called?

- ☐ Particle energy
- ☐ Enthalpy
- ☐ Activation energy
- ☐ Reaction energy
- ☐ Heat energy

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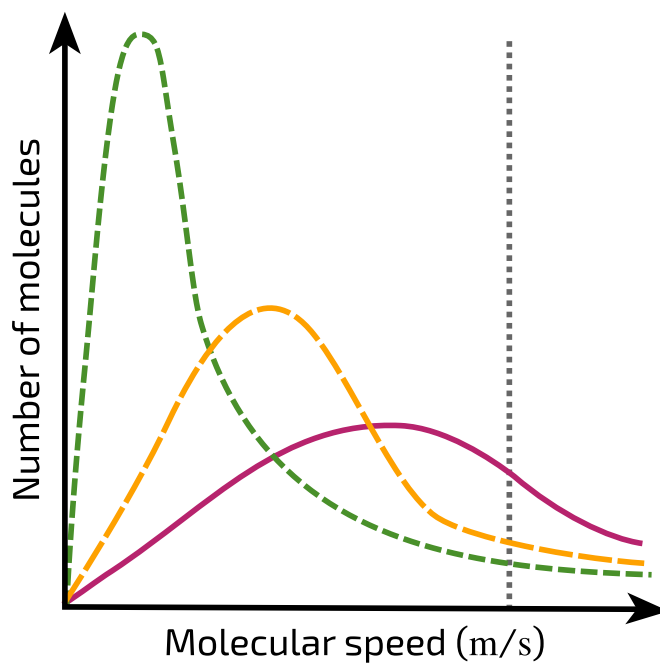
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## Three Boltzmann Distributions

**Subject & topics:** Chemistry | Physical | Kinetics      **Stage & difficulty:** A Level P1

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In this question the effect of temperature on the Maxwell-Boltzmann distribution will be explored.



**Figure 1:** Maxwell-Boltzmann distributions at different temperatures

**Part A**

**Highest temperature**

Which line on the diagram represents the distribution at the highest temperature?

- ☐ The yellow (long dashed) line
- ☐ The green (short dashed) line
- ☐ The red (solid) line
- ☐ The grey (dotted) line

**Part B**

**Lowest temperature**

Which line represents the distribution at the lowest temperature?

- ☐ The yellow (long dashed) line
- ☐ The green (short dashed) line
- ☐ The red (solid) line
- ☐ The grey (dotted) line

### Part C

#### Activation energy

The grey dotted line represents the activation energy. How could the activation energy be lowered?

- ☐ By increasing the temperature
- ☐ By introducing a catalyst
- ☐ By lowering the energy of the reaction mixture
- ☐ By giving the particles more kinetic energy
- ☐ By increasing the concentrations of the reactants

### Part D

#### Lowering the activation energy

If the activation energy was lowered, which way would the vertical line move on the diagram?

- ☐ To the left
- ☐ To the right
- ☐ It would not move

**Part E**

**Net effect**

What would be the net effect of this shift?

- ☐ The temperature at which half of the particles have the required activation energy would increase
- ☐ A greater number of particles would have sufficient energy to chemically react
- ☐ There would be no change in the number of particles that would have enough energy to react
- ☐ Fewer particles would have sufficient energy to chemically react

**Part F**

**Low activation energy**

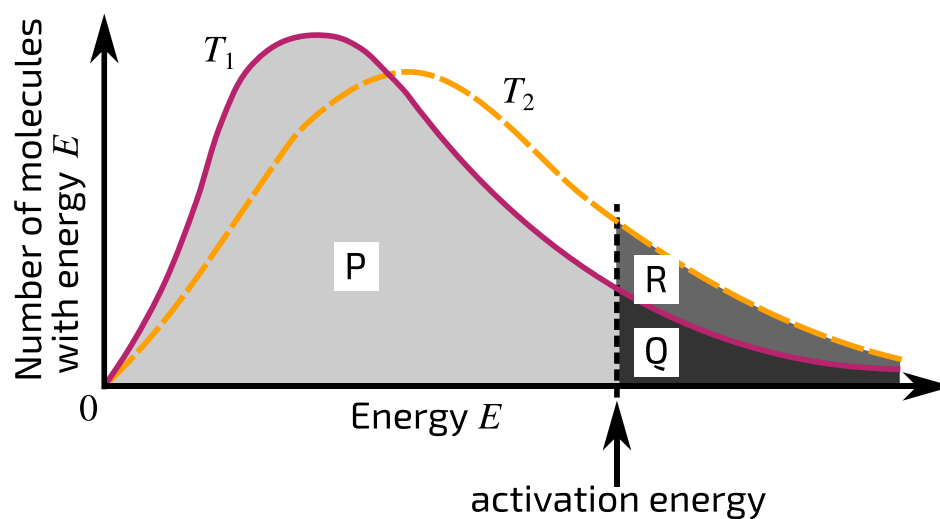
Imagine the activation energy was very low, so that the vertical line was on the left half of the diagram above. If a catalyst was then added, at which of the three temperatures would there be the greatest increase in the number of particles that have sufficient energy to react?

- ☐ The highest temperature
- ☐ The number of particles with sufficient energy would increase by the same amount for each of the three temperatures
- ☐ The middle temperature
- ☐ The lowest temperature

## Proportion Exceeding $E_a$

**Subject & topics:** Chemistry | Physical | Kinetics **Stage & difficulty:** A Level C1

The distribution of the number of molecules with energy  $E$  is given in the sketch for two temperatures,  $T_1$  and a higher temperature,  $T_2$ . The letters  $P$ ,  $Q$ ,  $R$  refer to the separate and differently shaded areas. The activation energy is marked on the energy axis.



**Figure 1:** Two Maxwell-Boltzmann distributions

Find an expression which gives the fraction of the molecules present which have at least the activation energy at the higher temperature  $T_2$ .

The following symbols may be useful:  $P$ ,  $Q$ ,  $R$

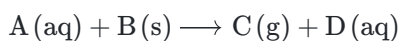
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## Rate of Gas Production

**Subject & topics:** Chemistry | Physical | Kinetics    **Stage & difficulty:** GCSE P3, A Level P1

The following table contains data collected from an experiment carried out to determine the rate of a general reaction, which can be represented by the following equation:



The volume of gas C was measured over time using a gas syringe.

| Time / seconds | Volume of gas produced / cm <sup>3</sup> |
|----------------|------------------------------------------|
| 0              | 0.0                                      |
| 5              | 1.2                                      |
| 10             | 1.9                                      |
| 15             | 2.4                                      |
| 20             | 2.7                                      |
| 25             | 2.9                                      |
| 30             | 3.0                                      |
| 35             | 3.1                                      |
| 40             | 3.1                                      |

### Part A

#### Reaction end

Using the data in the table, when did the reaction stop?



**Part B**

**Average rate**

What is the average rate of reaction from its start until the reaction's end?

**Part C**

**Increasing the rate**

What could be done to speed up the rate of this reaction from the following options?

- ☐ Decreasing the concentration of reactant A
- ☐ Decreasing the surface area of solid B
- ☐ Adding a catalyst
- ☐ Decreasing the temperature
- ☐ Introducing an inert solvent

Part D

Instantaneous rate

How would the instantaneous rate of reaction be determined?

- ☐ Plot the data, join up the points with straight lines and use the gradient of one of these straight lines to determine the rate at any time.
- ☐ Plot the data, draw a smooth line of best fit and draw a tangent at the required time. The gradient of this tangent is the instantaneous rate.
- ☐ Divide the total change in volume by the total change in time.
- ☐ Take the difference in volume between two data points, and divide by the difference in time.

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## Average Rate of Reaction

**Subject & topics:** Chemistry | Physical | Kinetics      **Stage & difficulty:** GCSE P3, A Level P1

An experiment was carried out by a student to determine the rate of a certain reaction, and how the rate changed over time. The reaction involved a solid being added to a solution, and the solid dissolving, subsequently producing a gas. The change in the mass of the reaction vessel was recorded over time to give information on the rate of reaction. A table of the data that was collected is given below.

| Time / seconds | Total mass of reaction vessel / g |
|----------------|-----------------------------------|
| 0              | 3.79                              |
| 10             | 3.22                              |
| 20             | 2.69                              |
| 30             | 2.16                              |
| 40             | 1.69                              |
| 50             | 1.35                              |
| 60             | 1.16                              |
| 70             | 1.06                              |
| 80             | 1.01                              |
| 90             | 0.99                              |
| 100            | 0.97                              |
| 110            | 0.95                              |
| 120            | 0.93                              |

**Part A**

**Average rate of reaction 1**

What is the average rate of the reaction up to 40 s? Give your answer in terms of the mass of gas produced.

**Part B**

**Average rate of reaction 2**

What is the average rate of reaction between 80 s and 120 s? Give your answer in terms of the mass of gas produced.

**Part C**

**Instantaneous rate**

By plotting a graph, calculate the instantaneous rate of reaction at 25 seconds. Give your answer in terms of the mass of gas produced.

**Part D**

**Instantaneous rate 2**

What is the instantaneous rate of reaction at 60 seconds? Give your answer in terms of the mass of gas produced.

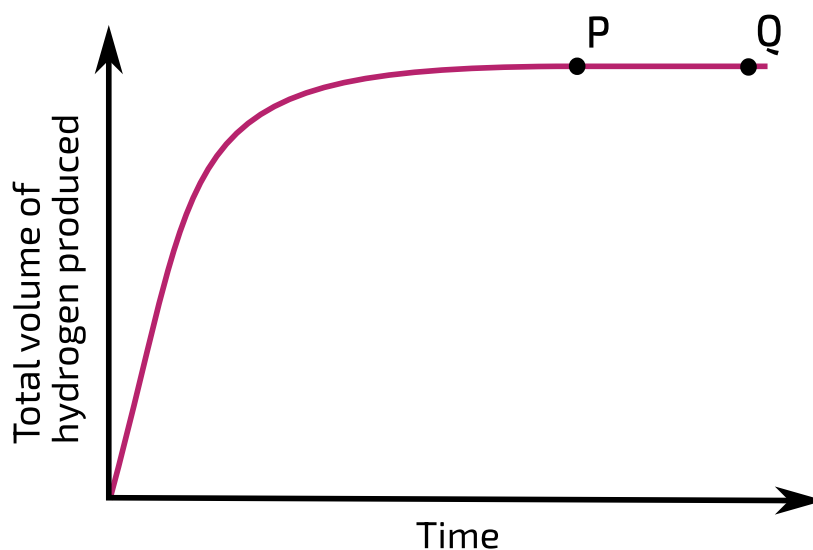
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## Rate of Hydrogen Production

**Subject & topics:** Chemistry | Physical | Kinetics    **Stage & difficulty:** GCSE P2, A Level P1

The graph below shows how the total volume of hydrogen, produced by the reaction between hydrochloric acid and an excess of magnesium, varies with time.



**Figure 1:** Graph showing the volume of hydrogen produced with time.

### Part A

#### Equation

Write the equation for this reaction, balancing to use the lowest integer coefficients possible. Include state symbols.

**Part B**

**PQ**

Which one of the following statements about the section PQ of the curve is correct?

- ☐ The reaction is continuing at a constant rate.
- ☐ The rate of reaction is increasing.
- ☐ No more hydrogen is being produced.
- ☐ All the magnesium has reacted.
- ☐ The rate of production of hydrogen is at a maximum.

**Part C**

**Reaction end**

The graph shows the reaction stopping at point P. Under what conditions could the reaction stop at point Q?

- ☐ Double the concentration of acid, and halve the volume used.
- ☐ Same amount of acid used, but with a smaller amount of magnesium used, so the acid was now significantly in excess.
- ☐ Same initial conditions, but with a catalyst added.
- ☐ Double the amount of acid used (the magnesium is still in excess).
- ☐ Same mass of magnesium, but with a smaller surface area.

Adapted with permission from UCLES, O Level Chemistry, June 1986, Paper 1, Question 9

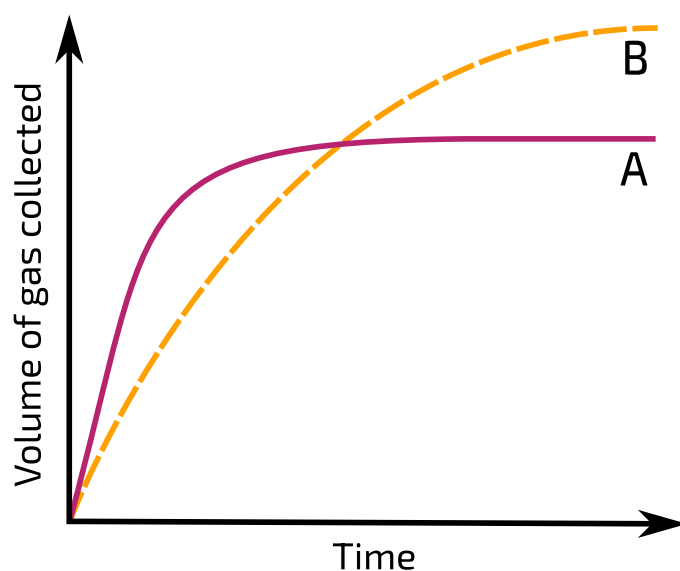
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## Interchanging Graphs

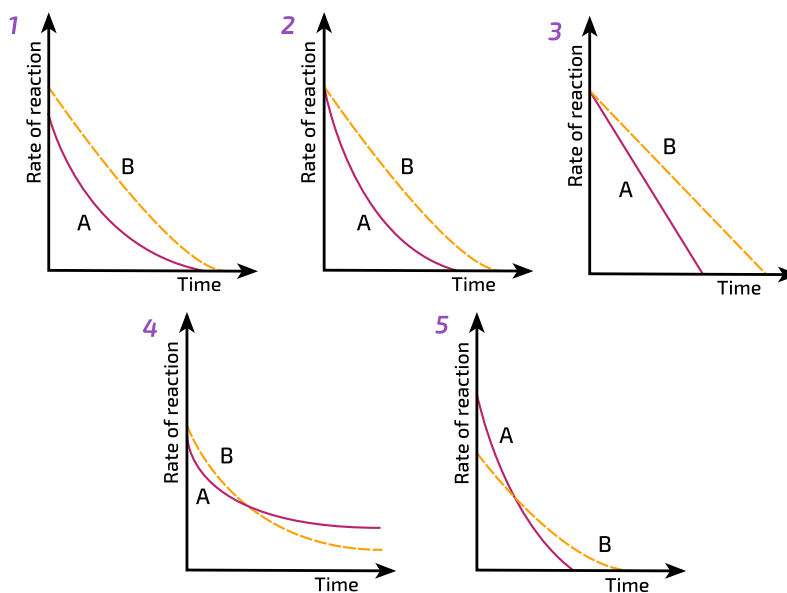
**Subject & topics:** Chemistry | Physical | Kinetics      **Stage & difficulty:** A Level P2

In two separate experiments, experiment **A** and experiment **B**, a substance was decomposed and the gas evolved was collected. The graph below shows the total volume of gas collected against time for each experiment.



**Figure 1:** Volume of gas collected against time

Below are some graphs showing possible rates of reaction against time for the two experiments.





**Figure 2:** The rate against time for the five different experiments.

Which one of the above graphs shows how the rate of reaction varied with time in the experiments A and B?

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5

Adapted with permission from UCLES, O Level Chemistry, June 1985, Paper 1, Question 17

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## Marble Chips

**Subject & topics:** Chemistry | Physical | Kinetics      **Stage & difficulty:** GCSE P3, A Level P1

Marble chips, assumed to be made up solely of  $\text{CaCO}_3$  (s) in this experiment, react with dilute hydrochloric acid according to the following reaction:



The rate of this reaction was investigated by measuring the volume of gas produced over time. 16 g of marble chips and an excess of dilute acid was used, and the experiment was carried out at 25 °C.

The results of the experiment are given in the table below.

| Time / seconds | Volume of gas produced / $\text{dm}^3$ |
|----------------|----------------------------------------|
| 0              | 0.00                                   |
| 10             | 0.23                                   |
| 20             | 0.35                                   |
| 30             | 0.44                                   |
| 40             | 0.51                                   |
| 50             | 0.56                                   |
| 60             | 0.60                                   |
| 70             | 0.63                                   |
| 80             | 0.64                                   |
| 90             | 0.64                                   |

### Part A

#### Plotting the graph

Plot a graph of the data, and draw a smooth line of best fit.

How would the initial gradient of the graph change if the reaction was instead carried out at  $40^{\circ}\text{C}$ ?

- ☐ The initial gradient would be smaller.
- ☐ The initial gradient would be greater.
- ☐ More information is needed to answer this.
- ☐ The initial gradient would not change.
- ☐ The initial gradient would be zero, as the reaction would not proceed at this temperature.

### Part B

#### Initial gradient

How would the initial gradient change if a single piece of marble of mass 16 g was used instead of many smaller marble chips with total mass 16 g?

- ☐ More information is needed to answer this.
- ☐ The initial gradient would be smaller.
- ☐ The initial gradient would not change.
- ☐ The initial gradient would be greater.
- ☐ The initial gradient would be zero, as the reaction would not proceed.

**Part C**

**Collision theory**

In terms of particles, why would the rate of reaction increase if the reaction were carried out with concentrated, instead of dilute, hydrochloric acid?

**Part D**

**Average rate of reaction**

Use your graph to calculate the average rate of reaction over the first 60 seconds.

**Part E**

**Instantaneous rate of reaction**

Use your graph to determine the rate of reaction at 50 seconds (the instantaneous rate). Give your answer to 1 significant figure.

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